



UNIVERSITY OF WESTERN AUSTRALIA

MASTER OF PHYSICS (COMPUTATIONAL PHYSICS)

**Quantum Circuits for Solving Differential Equations via
Physics-Informed Effective Hamiltonians and GQSP
Quantum Imaginary Time Evolution**

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Supervised by

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16 September 2026

Declaration

This is to certify that:

1. This thesis comprises of my original work.
2. Due acknowledgment has been made in the text to all other materials used.
3. This thesis consists of less than 60 pages inclusive of tables and figures but exclusive of references and appendices.

I authorise the Head of the Department of Physics to make a copy of this thesis to any person judged to have an acceptable reason for access to the information, i.e, for research, study or instruction.



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Summary of Student Achievement

Abstract

Brief summary.....

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Notation and Conventions

A quick reference for symbols used across the thesis; each is properly introduced at its first use in the text below.

Discretisation

Symbol	Meaning
n	Number of qubits in the Chebyshev coefficient register (the resolution).
$N = 2^n$	Chebyshev truncation degree; dimension of the single-field Hilbert space.

Differential-equation structure

Symbol	Meaning
k	Order of an ordinary differential equation (ODE).
p	Polynomial (nonlinear) degree of the ODE (§4).

Chebyshev and descriptor operators

Symbol	Meaning
T_ℓ	Chebyshev polynomial of the first kind, degree ℓ .
G	Dense $N \times N$ Chebyshev differentiation matrix (the collocation route’s operator; Wu et al. [1]).
$\tilde{B}, \tilde{D}$	Banded factors of G (Appendix B).
M	Coefficient-multiplication operator in Chebyshev coefficient space (Appendix B).
w	The stacked descriptor state, $w = (c^{(0)}; \dots; c^{(k)})$.
A	The residual operator of a linear-system reformulation of the ODE (dense in the collocation route, banded in the descriptor route).

Physical and complexity quantities

Symbol	Meaning
H	The physics-informed effective Hamiltonian (Wu et al. [1]).
α	Block-encoding subnormalisation factor.
Δ	Spectral gap of H between its ground state and first excited state.
ϵ_G	Block-encoding / truncation error associated with representing G .

1 Introduction

1.1 Differential Equations as a Target for Quantum Computation

1.2 The Encoding Wall in Physics-Informed Quantum Solvers

1.3 Contributions and Scope

1.4 Outline

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2 Background and Literature Review

2.1 Quantum Computation Preliminaries

In classical computing, the fundamental unit of information is the bit, which can take one of two values: 0 or 1.

The fundamental unit of information in quantum computing is the qubit, which can exist in a superposition of states, allowing it to encode both 0 and 1 simultaneously. This property enables quantum computers to perform certain computations more efficiently than classical computers.

Definition 1 (State Space): The set of all quantum states that a quantum system can occupy is a complex vector space with inner product structure known as a Hilbert space (denoted $\mathcal{H}$). A finite dimensional Hilbert space is isomorphic to $\mathbb{C}^N$ ($\mathcal{H} \cong \mathbb{C}^N$) for some $N \in \mathbb{N}$, where N is the dimension of the Hilbert space. A quantum state is a unit vector in this Hilbert space, represented in dirac notation as $|\psi\rangle \in \mathcal{H}$, with $\langle\psi|\psi\rangle = 1$ (normalisation condition). A quantum state $|\psi\rangle$ can be expressed as a linear combination of basis states $\{|i\rangle\}_{i=0}^{N-1}$, where $|i\rangle$ is the i^{th} basis state of the Hilbert space.

$$|\psi\rangle = \sum_{i=0}^{N-1} \psi_i |i\rangle = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \psi_{N-1} \end{pmatrix} \quad (2.1)$$

Its hermitian conjugate (or adjoint) is denoted as $\langle\psi| = |\psi\rangle^\dagger$. Note that for some $e^{i\theta}$ for $\theta \in [0, 2\pi)$, the states $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ are physically indistinguishable, and thus correspond to the same physical quantum state. This is known as a global phase factor.

Definition 2 (The Qubit): A quantum bit (qubit) is a two-state quantum system, characterised by $|\psi\rangle \in \mathbb{C}^2$.

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad (2.2)$$

where $\alpha, \beta \in \mathbb{C}$, $|0\rangle$ and $|1\rangle$ are the computational basis states, and $|\alpha|^2, |\beta|^2$ are the probabilities of measuring the qubit in the $|0\rangle$ and $|1\rangle$ states, respectively. The normalisation condition requires that $|\alpha|^2 + |\beta|^2 = 1$.

Definition 3 (Composite Systems): The state space of a composite quantum system is the tensor product of the state spaces of its subsystems. Consider two quantum states $|\psi\rangle_A \in \mathcal{H}_A$

and $|\psi\rangle_B \in \mathcal{H}_B$, the composite state $|\psi\rangle_{AB}$ is given by:

$$|\psi\rangle_{AB} = |\psi\rangle_A \otimes |\psi\rangle_B \in \mathcal{H}_A \otimes \mathcal{H}_B \quad (2.3)$$

where $\otimes$ denotes the tensor product and is often hidden by convention. The dimension of the composite state space is the product of the dimensions of the individual state spaces. If a composite system cannot be written as a tensor product of its subsystems, it is said to be entangled. Entanglement is a unique feature of quantum mechanics that has no classical counterpart and plays a crucial role in quantum information processing.

We often consider the state of a composite system of n qubits, which is represented in a 2^n -dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^{2^n}$.

2.1.1 Quantum Gates and Circuits

Definition 4 (Quantum Algorithm): A quantum algorithm is a sequence of operations that manipulates qubits to perform a specific computation.

The goal of any quantum algorithm is to manipulate quantum states in a controlled manner to achieve a desired outcome. This is accomplished through the application of quantum gates, which are unitary operations that transform quantum states. Quantum gates can be combined to form quantum circuits, which represent the sequence of operations performed on qubits. Each quantum gate can be implemented using physical operations on hardware, and the design of efficient quantum algorithms is a central challenge in quantum computing research.

Definition 5 (Unitary Evolution): The evolution of a closed quantum system between two states is described by unitary transformations $U \in \mathbb{C}^{2^n \times 2^n}$, where n is the number of qubits in the system.

$$|\psi\rangle_{final} = U |\psi\rangle_{initial}, \quad U^\dagger U = U U^\dagger = I \quad (2.4)$$

where $U^\dagger$ is the hermitian conjugate of U and I is the identity operator. Unitary transformations preserve the inner product and thus the norm of quantum states, ensuring that states have a physical interpretation as probability amplitudes.

These unitary transformations are known as quantum gates, and each gate serves a specific purpose in manipulating quantum states.

The quantum circuit diagram offers a succinct and intuitive representation of quantum algorithms, analogous to Feynman diagrams in quantum field theory.

Time flows left to right, and each wire (horizontal line) represents a qubit or a quantum register. Gates are placed sequentially on the wires indicating operations applied to qubits. An example is the Hadamard gate, which creates superposition, and the CNOT gate, which entangles qubits. A full list of quantum gates and their corresponding circuit symbols can be found in .

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2.1.2 Complexity of Quantum Algorithms

Definition 6 (Gate Complexity): The total number of quantum gates used in a quantum circuit.

Definition 7 (Circuit Depth): The maximum number of gates applied sequentially on any qubit in a quantum circuit.

Definition 8 (Ancillary Qubits): Additional qubits introduced into a quantum circuit beyond the primary input qubits, typically initialized to a baseline state (such as $|0\rangle$) to assist with intermediate calculations, temporary storage, or entanglement generation.

Definition 9 (Big- $\mathcal{O}$ Notation): Big- $\mathcal{O}$ notation is a mathematical notation used to describe the asymptotic behavior of functions.

$$\mathcal{O}(f(n)) = \{g(n) : \exists c > 0, n_0 \in \mathbb{N} \text{ such that } 0 \leq g(n) \leq cf(n) \text{ for all } n \geq n_0\} \quad (2.5)$$

In the context of quantum algorithms, it is used to express the scaling of gate complexity, circuit depth, and space requirements with respect to the number of qubits n .

For example, if a quantum algorithm has a gate complexity of $\mathcal{O}(n^2)$, it means that the number of gates required grows quadratically with the number of qubits.

A quantum circuit is said to be **efficient** if its circuit depth, gate complexity, and the number of ancillary qubits all scale polynomially with the number of qubits n .

2.1.3 Noisy Intermediate-Scale Quantum (NISQ) Era

Write this section. Use Josh's as a guide.

2.2 Quantum Algorithms for Differential Equations

2.2.1 Linear-Systems and Hamiltonian-Simulation Routes

2.2.2 Variational and Carleman-Linearisation Routes

2.2.3 Critical Assessment

2.3 Classical Spectral Methods and the Chebyshev Basis

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2.6 Readout and Amplitude Amplification

2.7 Synthesis: The Gap This Thesis Addresses

3 The Physics-Informed Hamiltonian Method

3.1 Overview

This chapter recreates the physics-informed effective Hamiltonian framework of [1] exactly as published, then subjects it to the same numerical scrutiny this thesis applies to its own contribution. The framework encodes a differential equation’s solution as the ground state of a residual Hamiltonian built from a dense Chebyshev differentiation matrix, via a construction reproduced here without modification. Section 3.2 states that construction in full; Sections 3.3 and 3.4 give the published encoding circuit and this thesis’s own reconstruction, verification and measurement of it; Section 3.5 diagnoses where the construction fails and motivates the descriptor reformulation of Section 4.3.

3.2 The Residual Construction and Its Ground State

Wu et al. recast solving a differential equation as three linked moves [1]: represent a candidate solution as a quantum state carrying its Chebyshev coefficients; write the differential equation and every boundary or initial condition as linear operators that must annihilate that state; and assemble a Hamiltonian from those operators whose ground state is exactly the state satisfying all of them at once.

The Chebyshev feature map. A solution f is approximated by a degree- $(N-1)$ truncated Chebyshev series, $N = 2^n$, carried by an n -qubit state

$$|\tau(x)\rangle_n = \frac{1}{2^{n/2}} T_0(x) |0\rangle^{\otimes n} + \frac{1}{2^{(n-1)/2}} \sum_{k=1}^{N-1} T_k(x) |k\rangle, \quad (3.1)$$

so that a coefficient vector $|\psi\rangle_n \in \mathbb{C}^N$ (unit norm, $\langle\psi|\psi\rangle = 1$) represents the model

$$f_Q(x) = \sqrt{\eta} \langle\tau(x)|\psi\rangle_n \quad (3.2)$$

for some positive scale η . Separating the *direction* $|\psi\rangle$ from the *scale* $\sqrt{\eta}$ this way matters because everything from here to the ground-state eigenvalue problem below depends only on $|\psi\rangle$: η cancels out of every constraint $A|\psi\rangle = 0$ and out of the Hamiltonian built from such constraints, so it plays no role until Equation (3.8) fixes it, once, from a single known data point, entirely after the ground state has been found. Every equation from here on is written for $|\psi\rangle$ alone, with η suppressed accordingly.

The derivative operator. Differentiating a Chebyshev series produces another Chebyshev series whose coefficients are a fixed linear function of the original ones [2]: writing this map in the normalised basis of Equation (3.1) gives an $N \times N$ matrix G with

$$G_{k,p} = \begin{cases} 2p, & p > k \geq 1, \text{ } p-k \text{ odd,} \\ \sqrt{2}p, & k = 0, \text{ } p \text{ odd, } p \geq 1, \\ 0, & \text{otherwise,} \end{cases} \quad (3.3)$$

so that $f'_Q(x) = \langle\tau(x)|G^\top|\psi\rangle_n$, and the m -th derivative follows from the power $G^{\top m}$. Equation (3.3) is dense and upper triangular: row k is non-zero at every column $p > k$ of the right parity, so the lowest-order coefficient can depend on as many as half of the others, and G carries $\Theta(N^2)$ non-zero entries in total. Appendix B gives this fill-in explicitly at $N = 4$, alongside the banded factors $\tilde{B}, \tilde{D}$ that Section 4.3 builds from the same recurrence.

The differential-equation row. For a linear equation of order k with constant coefficients, $\sum_{j=0}^k a_j f^{(j)} = 0$, direct substitution of $G^{\top j}$ for each derivative gives

$$A = \sum_{j=0}^k a_j G^{\top j}, \quad (3.4)$$

Wu et al.'s own construction [1]; a second-order equation, for instance, gives $A = a_2 G^{\top 2} + a_1 G^\top + a_0 I$ directly.

A coefficient $a_j(x)$ that varies with x cannot be substituted this way: $a_j(x) f^{(j)}(x)$ is a product of two functions, and the Chebyshev coefficients of a product are not the product of the coefficients.

Wu et al. resolve this with a second operator, the lifting operator M_{x^p} , built from the general Chebyshev product-to-sum identity

$$x^p T_k(x) = 2^{-p} \sum_{i=0}^p \binom{p}{i} T_{|k+p-2i|}(x), \quad (3.5)$$

the p -fold iterate of the pairwise rule $T_j T_k = \frac{1}{2}(T_{j+k} + T_{|j-k|})$ that reappears below in the nonlinear extension. Conjugating Equation (3.5) into the normalised bases of Equation (3.1) gives an operator M_{x^p} satisfying

$$x^p \langle \tau(x) |_n = \langle \tau(x) |_{n+1} M_{x^p} \quad (3.6)$$

(entries: Appendix C). Multiplying a degree- $(N-1)$ model by x^p can raise its degree by as much as p , so M_{x^p} is exact and rectangular, $2N \times N$: no truncation is needed once the result is represented in the one-qubit-wider register Equation (3.6) already targets. A general coefficient, expanded to a finite power series $a_j(x) = \sum_p c_{j,p} x^p$ (a Maclaurin truncation, for a non-polynomial coefficient), lifts by linearity to $M_{a_j} = \sum_p c_{j,p} M_{x^p}$, and the differential-equation row for the general, variable-coefficient case follows by composing M_{a_j} and $G^{\top j}$ term-by-term,

$$A = \sum_{j=0}^k M_{a_j} G^{\top j}, \quad (3.7)$$

with $M_{a_j} = a_j I$ recovering Equation (3.4) whenever a_j is constant.

Data constraints. Boundary and initial conditions are built from one rank-one point-evaluation generator, $B_n(x) = \sqrt{2^n} |0\rangle^{\otimes n} \langle \tau(x) |_n$, non-zero only in its first row. A condition with a zero target is *invariant* – unchanged under $|\psi\rangle \rightarrow c|\psi\rangle$ – and enters the Hamiltonian directly as one further linear constraint, $B|\psi\rangle = 0$, e.g. a Dirichlet condition $f(x_z) = 0$ giving $B_n(x_z)|\psi\rangle = 0$ and a Neumann condition $f'(x_m) = 0$ giving $B_n(x_m)G^{\top}|\psi\rangle = 0$. A condition with a non-zero target, $f(x_s) = y_s$, is not scale-covariant and instead fixes the amplitude $\sqrt{\eta}$ of Equation (3.2) directly, after the ground state has been found:

$$\sqrt{\eta} = \frac{y_s}{\langle \tau(x_s) | \psi_g \rangle_n}. \quad (3.8)$$

Only one such *regular* constraint is used per solve, and Equation (3.8) is the only place η re-enters the pipeline.

Definition 10 (Physics-informed effective Hamiltonian): Collecting the differential-equation row A and every invariant data constraint B_i ,

$$H = A^{\top} A + \sum_i B_i^{\top} B_i, \quad \langle \psi | H | \psi \rangle = \|A\psi\|^2 + \sum_i \|B_i\psi\|^2. \quad (3.9)$$

H is Hermitian and positive semi-definite, so $\langle \psi | H | \psi \rangle \geq 0$ with equality exactly when every constraint holds simultaneously. The differential equation problem is thereby an eigenvalue problem, $H |\psi_g\rangle = \lambda_{\min} |\psi_g\rangle$, $\lambda_{\min} \approx 0$, whose ground state $|\psi_g\rangle$ is the solution.

Forming $A^\top A$ squares A 's own condition number, so H is markedly more ill-conditioned than A itself, and its ground state sits at the base of a highly compressed spectrum. Section 3.5 traces this to the measured spectral-gap collapse of Section 3.4.

The nonlinear extension. Wu et al. extend Definition 10 to polynomial nonlinearity by doubling the Hilbert space, $|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle$, and folding a product of two coefficient vectors back into one register via the same product-to-sum identity, $T_j T_k = \frac{1}{2}(T_{j+k} + T_{|j-k|})$ [1]. A residual quadratic in $|\psi\rangle$ of this kind annihilates a subspace of the $2N$ -dimensional $|\Psi\rangle$ far larger than the single physical direction: Wu et al. themselves note that the resulting H carries *at least* 2^{2n-1} degenerate zero-energy states, half of the doubled Hilbert space at minimum. No imaginary-time filter can select a direction inside a degenerate zero-energy cluster, so extracting the physical solution from this construction is classical post-selection within a known kernel rather than genuine state preparation; Section 5 addresses this, for a restricted but practically important class of problems, with a different construction.

3.3 The Published Collocation Encoding

Realising H as a circuit requires a unitary U_H that block-encodes it, $\langle 0 |_a U_H | 0 \rangle_a = H/\alpha$ for some $\alpha \geq \|H\|$, so that ground-state preparation and readout can act on U_H without reference to how H was built. The published route encodes A , and hence H , exactly as Section 3.2 assembles it: as a dense operator, with no attempt to expose or exploit any further structure.

A block encoding of a general dense matrix costs at least as many data-loading gates as the matrix has distinct entries that must be individually addressed, realised as a rotation controlled on the matrix's row and column indices [3], [4]. $G^\top$ is not sparse in this sense: by Equation (3.3), column p of $G^\top$ takes only the two values $\{2p, \sqrt{2}p\}$, but which of N possible columns is being addressed still varies from entry to entry, so a column of $G^\top$ still requires $\Theta(N)$ distinct addressed values rather than the $O(1)$ a banded or Toeplitz structure would permit.

Wu et al.'s own circuit avoids an exhaustive $\Theta(N)$ -branch multiplexed rotation by loading this near-linear dependence on the column index approximately, via a ladder of n singly-controlled rotations on one flag qubit [1]. Because rotations on the same qubit compose by exact angle addition, the ladder's net rotation for column index p is a sine of a sum of arcsines of p 's bits, which only approximates the desired value $2p$ – sine of a sum is not the sum of sines – with an error that shrinks rapidly as the per-bit angle shrinks.

This thesis reconstructs that circuit gate-for-gate from the source paper's own description, since its precise two-ancilla wiring is not fully recoverable from the published figure at the resolution available, and verifies the reconstruction bit-exact against an independently built

Quantikz circuit: the n -qubit column-index register, one wire per bit controlling an $R_y(\theta_m)$ gate on the flag qubit; the row-zero flag and its correction ladder; and the ripple-carry adder/uncompute wrapping the register.

Figure 1: **The reconstructed encoding circuit for $G^\top$'s column-linear entries.** A ripple-carry adder shifts the n -qubit column-index register into the form the ladder consumes, and is uncomputed afterwards. A ladder of n singly-controlled $R_y(\theta_m)$ rotations accumulates the amplitude $\text{raw}(p)$ of Equation (D.2) on one flag qubit by exact angle addition; a second, independently controlled ladder applies the row-zero correction. Ancilla: $n + 4$ total (Appendix D).

exact register-arithmetic alternative – an $O(2^n)$ -depth circuit with no approximation at all, kept purely as a verification reference – for n up to 5. Where the source paper states only that its errors are “negligible” and decrease “exponentially” with n , Appendix D derives a rigorous closed-form entrywise error bound from the same small-angle mechanism and verifies it against the measured error at every n tested from 2 to 8: the bound falls by roughly a further order of magnitude with each additional qubit, from 1.9×10^{-2} at $n = 2$ to 1.4×10^{-9} at $n = 8$ (Table 1), making the qualitative published claim quantitative.

The reproduced circuit uses $n + 4$ ancilla qubits (the exact register-arithmetic alternative needs $n + 5$), within one qubit of the $n + 3$ resource count reported in the source paper's own figure – Appendix D attributes the discrepancy to this reconstruction keeping two flag qubits live where the published circuit appears to combine them into one. Composing this encoding of G into A , and A into H via Equation (3.9), follows the standard linear-combination-of-unitaries and Gram-squaring algebra common to every block encoding built in this thesis: forming a Gram matrix from an encoded operator roughly doubles its ancilla count and squares its subnormalisation, and summing several such terms for the differential-equation row and every boundary condition adds their individual weights. Section 3.4 reports what this composition costs once carried through to the fully assembled H .

3.4 Numerical Verification: Reproducing and Characterising the Published Encoding

Verification proceeds in two stages, each checked against an independent classical reference so that a construction error in H cannot be mistaken for a state-preparation error downstream. First, encoding exactness: every block-encoded atom – G , each M_{x^p} , each boundary-condition

generator – is verified column-by-column against its reference matrix to 10^{-9} – 10^{-11} , and the Gram-and-sum recipe that combines them into H is verified to $< 10^{-10}$ on small test cases. Second, state preparation: the prepared ground state is compared to the classical `eigh` ground state of the same assembled H by cosine similarity, with that classical target itself cross-checked against an independent reference solver, separating a construction error in H from a filter-convergence error in the preparation stage. For the nonlinear extension of Section 3.2, the same protocol measures the degenerate kernel directly: null spaces occupy as much as 98% of $\dim(H)$ on the cases tested, well past the 2^{2n-1} floor Wu et al. state.

The composed H is not verified end-to-end – at the qubit counts the headline cases in Section 6 require, a dense classical simulation would need hundreds of gigabytes and is not attempted – so correctness beyond a handful of qubits rests on this compositional argument rather than a direct check of the assembled operator. Within that scope, three measurements characterise the published route empirically rather than merely reproduce the source paper’s own qualitative claims.

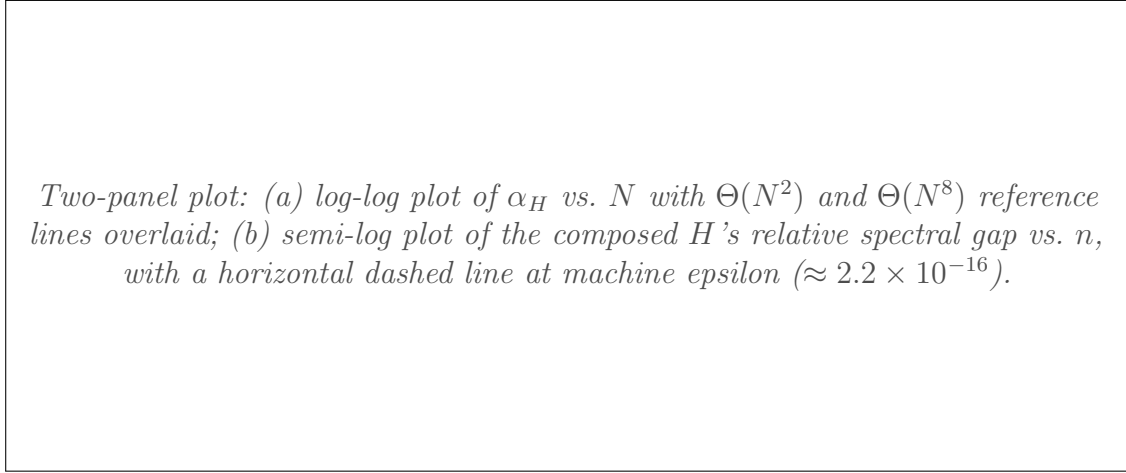


Figure 2: Measured subnormalisation and spectral-gap collapse of the composed H , $n = 2$ to 8. (a) α_H against $N = 2^n$ on log-log axes, empirically $\Theta(N^8)$, against $\|H\| = \Theta(N^2)$. (b) Relative spectral gap of the composed H against n , falling to $\sim 10^{-15}$ by $n \gtrsim 7$, below double-precision machine epsilon (dashed line).

First, conditioning: for the stiff pilot equation of Equation (3.4) (worked at $N = 4$ in Appendix C), the ground eigenvalue $\lambda_{\min}$ is not yet numerically zero at $n = 4$ ($\lambda_{\min}/\alpha \approx 3.9 \times 10^{-5}$, an under-resolved discretisation rather than a genuine kernel), but reaches $\lambda_{\min}/\alpha \approx 1.4 \times 10^{-14}$ by $n = 5$ – the same equation moving, within a single extra qubit, from an unreliable ground state to a numerically exact one. Second, subnormalisation: $G^\top$'s own S -norm subnormalisation was verified directly against `pihm.chebyshev_derivative_matrix` for $n = 1$ through 8 and grows empirically as $\Theta(N^4)$ (Appendix D); composing this into H via a Gram square inflates it to $\alpha_H \approx \Theta(N^8)$ against $\|H\| = \Theta(N^2)$, a subnormalisation tax that grows polynomially in N rather than staying tight, with a Haar-average post-selection probability that correspondingly falls like $N^{-3/2}$ rather than remaining constant. Third, and most consequential: the relative spectral gap of the fully assembled H , computed classically, falls to $\sim 10^{-15}$ by $n = 7$ – below

double-precision machine epsilon, so the ground state this Hamiltonian encodes is not numerically resolvable at all at that resolution, independent of how it is later block-encoded into a circuit. Section 6 reports the full per-case measurement and cost table.

3.5 Critical Assessment: The Encoding Wall

Three separate measurements in Section 3.4 point at the same structural cause. G 's $\Theta(N)$ distinct column values force any dense encoding of it to pay a data-loading cost that scales with N rather than with $\log N$; squaring A into $H = A^\top A$ squares that cost again in the subnormalisation; and summing the differential-equation row against every boundary-condition row compounds it further. The result is a block-encoded H whose subnormalisation grows like $\Theta(N^8)$ against an operator norm of only $\Theta(N^2)$ – a construction that, per query, is exponentially worse than it needs to be in the qubit count n , and polynomially worse than it needs to be in the Chebyshev resolution N .

This is not merely a cost inefficiency. Ground-state preparation by imaginary-time filtering needs a number of block-encoding queries that scales with α_H/Δ , where Δ is the spectral gap [5]; for any block encoding of this particular H , the gap itself is $\Delta = \Theta(1)$ against $\|H\| = \Theta(N^2)$, so $\alpha_H \geq \|H\|$ already forces at least $\tilde{\Theta}(N^2)$ queries before a single gate of the encoding is counted – neither route examined in this thesis is poly-logarithmic end to end (Section 7 completes this argument). The published collocation route misses even this floor by a further factor of $\approx N^6$, and the deficit is not merely asymptotic: at $n \gtrsim 7$, the composed H 's relative spectral gap has collapsed below double-precision machine epsilon, so the ground state it encodes cannot be resolved in simulation at any cost, let alone prepared efficiently on a quantum computer.

This chapter has reproduced the physics-informed effective Hamiltonian framework of Wu et al. exactly as published; the diagnosis above is a property of that published construction, not of any choice made in reproducing it. What makes A dense is the same thing that makes it hard to encode: G substitutes each derivative directly for the coefficients of the function it differentiates, in one step, and that substitution is exactly what a well-conditioned, banded alternative would decline to take. Section 4.3 develops that alternative, carrying the intermediate derivatives G eliminates as their own unknowns rather than substituting them away, so that the whole encoding-and-preparation pipeline inherits a structure the published route never had access to.

4 The Descriptor Reformulation

4.1 Overview

4.2 Design Rationale

The differentiation operator G is dense. Generally, dense operators are expensive to block-encode and sparse matrices are cheap.

The doubled Hilbert space method can create degenerate kernels. Quantum imaginary time evolution (QITE) can't select the physical solution from amongst a set of degenerate ground states, so although it is possible to encode the solution to the original problem in the ground state of the Hamiltonian, it is not possible to extract it from the ground state of the descriptor Hamiltonian.

4.3 Descriptor Form Construction

Differentiation of a Chebyshev series is a sparse operation written densely. Let $f = \sum_{\ell} c_{\ell}^{(0)} T_{\ell}$ have derivative $f' = \sum_{\ell} c_{\ell}^{(1)} T_{\ell}$; for $\ell \geq 2$ the coefficient vectors $c^{(0)}$ and $c^{(1)}$ are tied by the recurrence

$$c_{\ell-1}^{(1)} - c_{\ell+1}^{(1)} = 2\ell c_{\ell}^{(0)}, \quad (4.1)$$

with the $\ell = 1$ row carrying an extra factor of 2 on $c_0^{(1)}$, a normalisation artefact of T_0 that Appendix B states in full and derives. Each row relates the coefficient $c_{\ell}^{(0)}$ to just two coefficients of the derivative. In the normalised Chebyshev basis this is a matrix identity

$$\tilde{B} c^{(1)} = \tilde{D} c^{(0)}, \quad (4.2)$$

with $\tilde{B}$ banded (its main diagonal and its second superdiagonal), and $\tilde{D}$ carrying a single superdiagonal, such that both hold $\mathcal{O}(N)$ non-zero entries. The dense Chebyshev derivative matrix G is the operator that sends a function's coefficients directly to those of its derivative, $c^{(1)} = G c^{(0)}$. A simple rearranging of Equation (4.2) recovers it as the inverse

$$G = \tilde{B}^{-1} \tilde{D}, \quad (4.3)$$

an identity that holds exactly. The density of G is due to inverting a banded matrix, and Equation (4.2) is the same linear map before that inverse is done. This is the discrete form of the ultraspherical spectral method [6], in which differentiation becomes sparse once the derivative is allowed to sit in a polynomial basis one order above the function.

Consider the homogeneous order- k linear ODE

$$\sum_{j=0}^k a_j(x) f^{(j)}(x) = 0, \quad x \in [-1, 1], \quad (4.4)$$

Add a reference for this

Remember to define dense vs sparse as part of background

where $a_j(x)$ are the (possibly x -dependent) coefficients of each j^{th} derivative $f^{(j)}$.

We discretise by representing every function appearing in the equation by its first N coefficients in the normalised Chebyshev basis. Concretely, let $c^{(j)} \in \mathbb{C}^N$ denote the coefficient vector of $f^{(j)}$, for $j = 0, \dots, k$ and stack these into a single vector,

$$w = (c^{(0)}; c^{(1)}; \dots; c^{(k)}). \quad (4.5)$$

Two kinds of linear relations couple the blocks $c^{(j)}$:

Recurrence rows. The coefficients of consecutive derivatives are related by

$$-\tilde{D} c^{(j)} + \tilde{B} c^{(j+1)} = 0, \quad j = 0, \dots, k-1. \quad (4.6)$$

This gives k rows of the system, each enforcing the derivative relationship between $c^{(j+1)}$ and $c^{(j)}$. These rows are algebraic since they carry no derivative and only fix each $c^{(j+1)}$ as the derivative of $c^{(j)}$, being Equation (4.2) applied block-wise and hence equivalent to $c^{(j+1)} = Gc^{(j)}$ on $\ker A$.

Differential row. The differential equation itself becomes one further row, obtained by multiplying each coefficient vector $c^{(j)}$ by the (possibly x -dependent) coefficient $a_j(x)$ and summing:

$$\sum_{j=0}^k M_{a_j} c^{(j)} = 0, \quad (4.7)$$

where M_{a_j} is the matrix representing multiplication by $a_j(x)$ in coefficient space (reducing to $a_j I_N$ when a_j is constant). This single row is the equation itself: we carry the intermediate derivatives as their own unknown coefficient vectors within w , rather than representing them via $c^{(j)} = G^j c^{(0)}$, so that Equation (4.3) (the only dense step) never appears.

Together these $k+1$ block rows (k recurrence rows and one differential row) define a linear system for w and transform the original ODE problem into a differential-algebraic equation (DAE) system.

Definition 11 (Descriptor form residual operator): Collecting all block rows, the descriptor form chebyshev residual operator A is the $(k+1)N \times (k+1)N$ block matrix

$$A = \begin{pmatrix} -\tilde{D} & \tilde{B} & & & \\ & -\tilde{D} & \tilde{B} & & \\ & & \ddots & \ddots & \\ & & & -\tilde{D} & \tilde{B} \\ M_{a_0} & M_{a_1} & \dots & \dots & M_{a_k} \end{pmatrix}, \quad (4.8)$$

where the first k rows are the recurrence rows and the $(k+1)$ -th row is the differential row, so

that the whole system is compactly written as

$$Aw = 0. \quad (4.9)$$

A first-order example. For $f' + 2f = 0$ the state is $w = (c^{(0)}; c^{(1)})$, and the two facts relating its blocks are the recurrence $\tilde{B}c^{(1)} - \tilde{D}c^{(0)} = 0$ and the equation $2c^{(0)} + c^{(1)} = 0$, giving

$$A = \begin{pmatrix} -\tilde{D} & \tilde{B} \\ 2I_N & I_N \end{pmatrix}, \quad w = \begin{pmatrix} c^{(0)} \\ c^{(1)} \end{pmatrix}. \quad (4.10)$$

Every block is $\tilde{B}$, $\tilde{D}$ or a multiple of the identity, so A is banded, against the dense $G + 2I$ that substituting $c^{(1)} = Gc^{(0)}$ would produce. The fully populated $N = 4$ instance of Equation (4.10), the entries of $\tilde{B}$ and $\tilde{D}$, and the exactness of Equation (4.3), are given in Appendix B.

Cost. The reformulation trades a higher dimensional Hilbert space (and with it a higher qubit count) for a sparser residual operator. w has $(k + 1)N$ components at the classical, unpadded level; realising the field register on $\lceil \log_2(k + 1) \rceil$ qubits pads this to $2^{\lceil \log_2(k + 1) \rceil} N$ (Appendix B), which is what should be compared against the N of the collocation form. Each block of A is $\tilde{B}$, $\tilde{D}$, or a banded multiplication operator (bandwidth q for a degree- q polynomial coefficient; Appendix B), so A carries $\mathcal{O}(1)$ occupied diagonals per block, whereas the collocation form residual $\sum_j a_j G^j$ is dense with $\Theta(N^2)$ non-zero entries. The $c^{(0)}$ -block of the recovered null vector is the Chebyshev coefficient vector the collocation form solves for, agreeing with it to within numerical precision, quantified in Section 6. Carrying the intermediate derivatives instead of eliminating them converts the dense physics-informed residual into a banded one, and every later stage of the pipeline inherits that structure.

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Source paper for this repo’s PIHM + qubitization + GQSP-QITE pipeline, 2025. arXiv: 2504.13174 [quant-ph].

- [2] L. N. Trefethen, *Spectral Methods in MATLAB*. SIAM, 2000.
- [3] C. Sünderhauf, E. Campbell and D. Camps, ‘Block-encoding structured matrices for data input in quantum computing,’ *Quantum*, vol. 8, p. 1226, 2024. arXiv: 2302.10949 [quant-ph].
- [4] D. Camps, R. Van Beeumen, L. Lin and C. Yang, ‘Explicit Quantum Circuits for Block Encodings of Certain Sparse Matrices,’ *SIAM Journal on Matrix Analysis and Applications*, vol. 45, no. 1, pp. 801–827, 2024. arXiv: 2203.10236 [quant-ph].
- [5] A. Gilyén, Y. Su, G. H. Low and N. Wiebe, ‘Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics,’ in *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing (STOC 2019)*, Cited in-repo only as “Gilyén et al. (2019)” with no arXiv ID; identity/eprint inferred, not stated in the citing text, 2019, pp. 193–204. arXiv: 1806.01838 [quant-ph].
- [6] S. Olver and A. Townsend, ‘A Fast and Well-Conditioned Spectral Method,’ *SIAM Review*, vol. 55, no. 3, pp. 462–489, 2013, Ultraspherical spectral method; banded, well-conditioned differentiation via a change of ultraspherical basis. Antecedent of the descriptor basis change $\tilde{B}$ (see arXiv:1202.1347 in References.md). DOI: 10.1137/120865458. arXiv: 1202.1347.
- [7] L. Lin, *Lecture notes on quantum algorithms for scientific computation*, <https://arxiv.org/abs/2201.08> Jan. 2022.
- [8] O. Kyriienko, ‘Quantum inverse iteration algorithm for programmable quantum simulators,’ *npj Quantum Information*, vol. 6, p. 7, 2020. arXiv: 1901.09988 [quant-ph].
- [9] S. Scali, C. Umeano and O. Kyriienko, *Quantum topological data analysis via the estimation of the density of states*, 2023. arXiv: 2312.07115 [quant-ph].
- [10] C. A. Williams, S. Scali, A. A. Gentile, D. Berger and O. Kyriienko, *Addressing the Readout Problem in Quantum Differential Equation Algorithms with Quantum Scientific Machine Learning*, Related work distinguished from this repo’s readout approach; not a competing technique, 2024. arXiv: 2411.14259 [quant-ph].
- [11] G. Brassard, P. Høyer, M. Mosca and A. Tapp, *Quantum Amplitude Amplification and Estimation*, 2000. arXiv: quant-ph/0005055.
- [12] D. W. Berry, A. M. Childs, R. Cleve, R. Kothari and R. D. Somma, ‘Exponential improvement in precision for simulating sparse Hamiltonians,’ in *Proceedings of the 46th Annual ACM Symposium on Theory of Computing (STOC 2014)*, Introduces the oblivious amplitude-amplification variant used in readout.py, 2014, pp. 283–292. arXiv: 1312.1414 [quant-ph].
- [13] Y. Suzuki, S. Uno, R. Raymond, T. Tanaka, T. Onodera and N. Yamamoto, ‘Amplitude estimation without phase estimation,’ *Quantum Information Processing*, vol. 19, p. 75, 2020. arXiv: 1904.10246 [quant-ph].

- [14] D. Grinko, J. Gacon, C. Zoufal and S. Woerner, ‘Iterative Quantum Amplitude Estimation,’ *npj Quantum Information*, vol. 7, p. 52, 2021. arXiv: 1912.05559 [quant-ph].
- [15] H. Nishi, T. Kosugi, Y. Huang, K. Hirose, Y. Okayama and Y.-i. Matsushita, *Quantum State Readout via Overlap-Based Feature Extraction*, Source of the per-basis-function SWITCH-test readout protocol specialised in readout.py, 2025. arXiv: 2505.08613 [quant-ph].
- [16] S. Chakraborty, R. Hazra, X. Li, C. Shao, Q. Wang and Z. Zhang, *Quantum singular value transformation without block encodings: near-optimal complexity with minimal ancilla*, Theorem A.1 gives the ancilla lower bound cited for exact block encodings of L-term decompositions, 2025. arXiv: 2504.02385 [quant-ph].
- [17] Z. Zhang and C. Shao, *Low-ancilla block encodings via Hamiltonian simulation*, Not yet implemented in this repo; flagged as the theoretically correct way to beat the current ancilla count, 2026. arXiv: 2607.01843 [quant-ph].
- [18] D. Motlagh and N. Wiebe, ‘Generalized Quantum Signal Processing,’ *PRX Quantum*, vol. 5, p. 020 368, 2024, The GQSP construction lib/GQSP/gqsp.py implements (Theorem 3). arXiv: 2308.01501 [quant-ph].
- [19] A. Klappenecker and M. Rötteler, ‘Discrete cosine transforms on quantum computers,’ *Proc. SPIE 4740, Image and Signal Processing and Analysis (ISPA 2001)*, 2001. arXiv: quant-ph/0111038.
- [20] *QRTlib*, Quantum real-transform (DCT/DST/Hartley) circuit library extending Klappenecker–Rötteler, 2025. arXiv: 2510.16625 [quant-ph].
- [21] *Dyadic-Order Quantum Fractional Transforms*, 2026. arXiv: 2604.09295 [quant-ph].
- [22] A. Barenco et al., ‘Elementary gates for quantum computation,’ *Physical Review A*, vol. 52, p. 3457, 1995.
- [23] D. Maslov, ‘Advantages of using relative-phase Toffoli gates with an application to multiple control Toffoli optimization,’ *Physical Review A*, vol. 93, p. 022 311, 2016. arXiv: 1508.03273 [quant-ph].
- [24] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th Anniversary. Cambridge University Press, 2010, Cited in-repo only as “eq. 5.5” with no edition/year given; edition/year here follow the standard published anniversary edition and are not independently confirmed against the citing text.
- [25] J. M. Martyn, Z. M. Rossi, A. K. Tan and I. L. Chuang, ‘Grand Unification of Quantum Algorithms,’ *PRX Quantum*, vol. 2, p. 040 203, 2021, Cited in-repo only as “Martyn, Rossi, Tan & Chuang (2021)” with no arXiv ID; identity/eprint inferred, not stated in the citing text. arXiv: 2105.02859 [quant-ph].
- [26] G. H. Low and I. L. Chuang, ‘Hamiltonian Simulation by Qubitization,’ *Quantum*, vol. 3, p. 163, 2019, Cited in-repo only as “Low & Chuang, seminal work” with no title/venue/arXiv ID; full citation inferred, not stated in the citing text. arXiv: 1610.06546 [quant-ph].

- [27] L. Grover and T. Rudolph, *Creating superpositions that correspond to efficiently integrable probability distributions*, Cited in-repo only as “Grover & Rudolph (2002)” with no arXiv ID; identity/eprint inferred, not stated in the citing text, 2002. arXiv: quant-ph/0208112.
- [28] M. Möttönen, J. J. Vartiainen, V. Bergholm and M. M. Salomaa, *Transformation of quantum states using uniformly controlled rotations*, Cited in-repo only as “Möttönen et al. (2004)” with no arXiv ID; identity/eprint inferred, not stated in the citing text, 2004. arXiv: quant-ph/0407010.
- [29] L. N. Trefethen, *Approximation Theory and Approximation Practice*. SIAM, 2013.
- [30] R. J. LeVeque, *Finite Difference Methods for Ordinary and Partial Differential Equations: Steady-State and Time-Dependent Problems*. SIAM, 2007.
- [31] J. P. Boyd, *Chebyshev and Fourier Spectral Methods*, 2nd. Dover Publications, 2001.
- [32] C. Canuto, M. Y. Hussaini, A. Quarteroni and T. A. Zang, *Spectral Methods: Fundamentals in Single Domains*. Springer, 2006, Used in this repo as the (independently consulted) stand-in for the source paper’s own “Voigt/Gottlieb/Hussaini” citation – see `voigtgottlieb1984spectral` and `References.md` item G31.
- [33] R. G. Voigt, D. Gottlieb and M. Y. Hussaini, *Spectral Methods for Partial Differential Equations*. SIAM, 1984, Title/year as named by the source paper’s (Wu et al. 2025, arXiv:2504.13174) own bibliography; not independently verified or consulted by this repo – see `References.md` item G31.
- [34] Qiskit contributors, *Qiskit: An Open-source Framework for Quantum Computing*, Primary quantum-circuit construction/simulation framework used throughout this repo; citation record added from the official Qiskit project, not transcribed from in-repo text, 2024. DOI: 10.5281/zenodo.2573505.
- [35] V. Bergholm et al., *PennyLane: Automatic differentiation of hybrid quantum-classical computations*, Used only in abandoned scratch experimentation (`notebooks/scratch/ResearchPlayground.ipynb`); not a live dependency of the canonical pipeline, 2018. arXiv: 1811.04968.

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A Quantum Gate Reference

B Chebyshev and Ultraspherical Identities

This appendix records the entries of the banded factors $\tilde{B}$ and $\tilde{D}$ introduced in Section 4.3, the exact factorisation $G = \tilde{B}^{-1}\tilde{D}$, and the fully populated $N = 4$ instance of the first-order descriptor operator.

B.1 The differentiation recurrence

For $f = \sum_{\ell \geq 0} a_\ell T_\ell$ with derivative $f' = \sum_{\ell \geq 0} p_\ell T_\ell$ on $[-1, 1]$, the Chebyshev coefficients satisfy

$$c_{\ell-1} p_{\ell-1} - p_{\ell+1} = 2\ell a_\ell, \quad \ell \geq 1, \quad (\text{B.1})$$

with $c_0 = 2$ and $c_\ell = 1$ for $\ell \geq 1$; the $\ell = 1$ row therefore carries the factor $c_0 = 2$ on p_0 . Truncating to $N = 2^n$ coefficients and dropping the p_N term, Equation (B.1) is the matrix identity $\tilde{B}_{\text{std}} p = \tilde{D}_{\text{std}} a$ with $\tilde{D}_{\text{std}}$ a single superdiagonal, $(\tilde{D}_{\text{std}})_{\ell, \ell+1} = 2(\ell + 1)$, and $\tilde{B}_{\text{std}}$ two diagonals, $(\tilde{B}_{\text{std}})_{\ell, \ell} = c_\ell$, $(\tilde{B}_{\text{std}})_{\ell, \ell+2} = -1$.

B.2 The normalised basis

The construction of Section 4.3 works in the normalised basis $\varphi_0 = T_0/2^{n/2}$, $\varphi_\ell = T_\ell/2^{(n-1)/2}$ ($\ell \geq 1$), orthonormal under the discrete N -point Chebyshev inner product $\langle T_k, T_\ell \rangle_N = \frac{N}{2} c_\ell \delta_{k\ell}$ [2]: since $N = 2^n$, this gives $\|\varphi_0\|_N = \|\varphi_\ell\|_N = 1$ exactly, so the coefficient vector of a unit-norm function is a unit vector. Writing $S = \text{diag}(\sqrt{c_\ell})$, the normalised factors are $\tilde{B} = S \tilde{B}_{\text{std}} S^{-1}$ and $\tilde{D} = S \tilde{D}_{\text{std}} S^{-1}$, that is

$$\tilde{D}_{\ell, \ell+1} = 2(\ell + 1)\sqrt{c_\ell/c_{\ell+1}}, \quad \tilde{B}_{\ell, \ell} = c_\ell, \quad \tilde{B}_{\ell, \ell+2} = -\sqrt{c_\ell/c_{\ell+2}}, \quad (\text{B.2})$$

so $\tilde{D}$ has the single non-zero diagonal $(2\sqrt{2}, 4, 6, 8, \dots)$ and $\tilde{B}$ has main diagonal $(2, 1, 1, \dots)$ and second superdiagonal $(-\sqrt{2}, -1, -1, \dots)$. Both hold $\mathcal{O}(N)$ non-zero entries at any N . The dense Chebyshev derivative matrix is their exact quotient,

$$G = \tilde{B}^{-1}\tilde{D}, \quad (\text{B.3})$$

an algebraic identity rather than a numerical approximation: $\tilde{B}$ and $\tilde{D}$ are read directly off the same recurrence that defines G (Equation (B.1)), so the equality holds exactly at every N , independent of whichever values a test suite happens to check. Appendix H reports this identity's residual against the reference implementation, alongside the rest of the composition-algebra checks.

B.3 The Ultraspherical Connection

The banded factorisation above is the first-derivative case of the ultraspherical spectral method of Olver and Townsend [6]. Stating the correspondence precisely is what fixes what this thesis adds to it.

Olver and Townsend represent a function's derivative not in the Chebyshev basis T_k but in the ultraspherical (Gegenbauer) basis $C_k^{(\lambda)}$, the family of polynomials orthogonal with respect to the weight $(1 - x^2)^{\lambda-1/2}$; at $\lambda = 1$ these are the Chebyshev polynomials of the second kind, $C_k^{(1)} = U_k$. Differentiation of a Chebyshev series obeys

$$\frac{dT_k}{dx} = k C_{k-1}^{(1)}, \quad k \geq 1 \quad (\text{B.4})$$

[6, Eq. 2.5], so for $u = \sum_k u_k T_k$ the coefficients of u' in the $C^{(1)}$ basis are read off directly: $(\mathcal{D}_0 u)_k = (k+1)u_{k+1}$, a single superdiagonal, sparse at every truncation. Because u' 's coefficients now live in a different basis to u 's own, a second, purely algebraic operator is needed to convert a function's own Chebyshev coefficients into the $C^{(1)}$ basis without differentiating it: $\mathcal{S}_0$, with $(\mathcal{S}_0)_{k,k} = 1$ ($k = 0$), $\frac{1}{2}$ ($k \geq 1$), and $(\mathcal{S}_0)_{k,k+2} = -\frac{1}{2}$ [6, Eq. 2.8]. Comparing entries, $\tilde{D}_{\text{std}} = 2\mathcal{D}_0$ and $\tilde{B}_{\text{std}} = 2\mathcal{S}_0$ exactly: the recurrence $\tilde{B}_{\text{std}} p = \tilde{D}_{\text{std}} a$ of Equation (B.1) is literally $\mathcal{S}_0 p = \mathcal{D}_0 a$, doubled by the convention that folds the weight of T_0 into $c_0 = 2$ rather than into the polynomial itself. This is the discrete shadow the framing of Section 4.3 claims: the descriptor recurrence does not compute a new identity, it asserts an existing one – that p 's own Chebyshev coefficients, converted into the $C^{(1)}$ basis by $\mathcal{S}_0$, must equal the derivative computed directly by $\mathcal{D}_0$ – as one row of a linear system, rather than solving it for p via $\mathcal{S}_0^{-1}$.

The generalisation to order $k \geq 1$ in [6] raises the basis one level per derivative: a λ -th derivative needs coefficients in $C^{(\lambda)}$, obtained from the analogous pair $dC_k^{(\lambda)}/dx = 2\lambda C_{k-1}^{(\lambda+1)}$ ($k \geq 1$, giving $\mathcal{D}_\lambda$, still a single superdiagonal) and a conversion operator $\mathcal{S}_\lambda : C^{(\lambda)} \rightarrow C^{(\lambda+1)}$, still two diagonals [6, Eqs. 3.2–3.3]. To assemble one equation of order k , every term is raised to the common top basis $C^{(k)}$ by a chain $\mathcal{S}_{k-1} \cdots \mathcal{S}_{\lambda+1}$ applied after each term's own $\mathcal{M}_\lambda[a^\lambda]\mathcal{D}_\lambda$, giving one almost-banded operator $L = \mathcal{M}_k[a^k]\mathcal{D}_k + \sum_{\lambda=0}^{k-1} \mathcal{S}_{k-1} \cdots \mathcal{S}_\lambda \mathcal{M}_\lambda[a^\lambda]\mathcal{D}_\lambda$ that is solved once for the coefficients of $u^{(k)}$ [6, §3.2] – so a k -th order equation needs k distinct conversion operators $\mathcal{S}_0, \dots, \mathcal{S}_{k-1}$ chained together and the lower-order unknowns eliminated.

The descriptor construction of Section 4.3 takes a different route to the same sparsity. Rather than raising the basis and eliminating, it keeps every derivative's coefficients in the *same* normalised Chebyshev basis, as its own block $c^{(j)}$ of w , and reuses the single $\lambda = 0$ pair $(\tilde{B}, \tilde{D})$ between every consecutive pair of blocks (Equation (4.6)). No elimination is performed, and no operator beyond $\tilde{B}$, $\tilde{D}$ and the multiplication operators of Appendix B.5 is required at any order k – the price is the larger state w that carries every intermediate derivative explicitly instead of eliminating it (Appendix B.5). What this thesis adds to [6] is not a new sparsification of differentiation – that is their result – but applying the *same* $\lambda = 0$ identity block-wise to build an unreduced, differential-algebraic residual for the physics-informed Hamiltonian of [1],

whose block structure is exactly what Section 4.4 block-encodes.

B.4 The $N = 4$ instance

At $n = 2$ ($N = 4$), Equation (B.2) gives

$$\tilde{B} = \begin{pmatrix} 2 & 0 & -\sqrt{2} & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \tilde{D} = \begin{pmatrix} 0 & 2\sqrt{2} & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad G = \begin{pmatrix} 0 & \sqrt{2} & 0 & 3\sqrt{2} \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{B.5})$$

The top-right entry $3\sqrt{2}$ of G is the fill-in: coefficient 0 of the derivative depends on coefficient 3 of the input. Row ℓ of G is non-zero at columns $\ell + 1, \ell + 3, \dots$, so this is the extreme case – the lowest-order coefficient can depend on up to half the others, while the highest-order one (row 3 above) depends on none. The factorisation removes the fill-in entirely: no row of $\tilde{B}$ or $\tilde{D}$ touches more than two coefficients.

For the first-order equation $f' + 2f = 0$ of Section 4.3, $w = (c^{(0)}; c^{(1)})$ and $A = \begin{pmatrix} -\tilde{D} & \tilde{B} \\ 2I_4 & I_4 \end{pmatrix}$ is the 8×8 matrix

$$A = \left[\begin{array}{cccc|cccc} 0 & -2\sqrt{2} & 0 & 0 & 2 & 0 & -\sqrt{2} & 0 \\ 0 & 0 & -4 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & -6 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \hline 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \end{array} \right]. \quad (\text{B.6})$$

The top four rows are the recurrence block $-\tilde{D}c^{(0)} + \tilde{B}c^{(1)} = 0$; the bottom four are the differential row $2c^{(0)} + c^{(1)} = 0$. Every non-zero entry lies on a diagonal or a near-diagonal, in place of the dense $G + 2I$ of the eliminated form.

B.5 Higher order, variable coefficients, and the padded block count

For an order- k equation the state is $w = (c^{(0)}; \dots; c^{(k)})$ and A stacks k recurrence block-rows $-\tilde{D}c^{(j)} + \tilde{B}c^{(j+1)} = 0$ above the single differential row $\sum_j M_{a_j} c^{(j)} = 0$: $k + 1$ block-rows, each N coefficients wide, giving the classical $(k + 1)N \times (k + 1)N$ square matrix of Definition 11.

A variable coefficient $a_j(x)$ enters through $M_{a_j} = \sum_q c_q M_{x^q}$, $0 \leq q \leq N$, where c_q are the *monomial* coefficients of a_j : supplied directly as an array $[c_0, c_1, \dots]$, or, when a_j is given as a function, obtained by a least-squares fit to degree $N - 1$ in the Chebyshev basis followed by an exact Chebyshev-to-monomial change of basis. The Chebyshev fit is what pins q to the same resolution N already fixed for the equation; its degree, not the subsequent change of basis, is

the only place a_j 's representation is truncated. M_{x^q} is the resolution- n , square multiplication-by- x^q operator built from $xT_\ell = \frac{1}{2}(T_{\ell+1} + T_{\ell-1})$ with the T_N term dropped. M_{x^q} is banded with bandwidth exactly q , so it folds into the same shift-and-diagonal form as $\tilde{B}$ and $\tilde{D}$. The dropped T_N term is the only approximation in the linear construction: for a smooth product resolved below degree $N - q$ the resulting error is confined to the top $\sim q$ Chebyshev modes and measures $\sim 10^{-13}$.

Padding to a power of two. The classical $(k+1)N \times (k+1)N$ matrix above is exact and complete as a linear system; it is not, however, the object any later stage of the pipeline actually builds. A gate-level realisation must index which block $c^{(j)}$ a given operator acts on with a qubit register, and a register of m qubits addresses 2^m blocks, not $k+1$ of them for an arbitrary k . A is therefore *embedded* in a $2^{\lceil \log_2(k+1) \rceil} N$ -dimensional space: the $k+1$ block-rows and block-columns above are kept exactly as derived, and $2^{\lceil \log_2(k+1) \rceil} - (k+1)$ further diagonal blocks are appended, each an identity I_N pinning its own dummy field to zero, $c_{\text{dummy}}^{(j)} = 0$. This padding is exact: the physical solution already satisfies the dummy rows trivially, since it has nothing in those blocks, so no error is introduced beyond the M_{x^q} truncation above. The qubit cost, however, is not uniform in k :

Order k	Blocks $k+1$	Field qubits $\lceil \log_2(k+1) \rceil$	Padded blocks $2^{\lceil \log_2(k+1) \rceil}$	Dummy blocks
1	2	1	2	0
2	3	2	4	1
3	4	2	4	0
4	5	3	8	3
7	8	3	8	0

so the state w the gate-level construction actually carries has $2^{\lceil \log_2(k+1) \rceil} N$ components, not $(k+1)N$: equal to the unpadded count exactly when $k+1$ is itself a power of two (orders 1, 3, 7, ...), and up to nearly double it otherwise – order 2, the most common case after order 1, already pays for a field register it only three-quarters uses. This is the qubit cost that must always accompany the gate-count comparison of Section 4.6.

C The Published Collocation Residual and the Lifting Operator M_{x^p}

This appendix gives the $N = 4$ instance of the collocation residual A that Section 3.2 builds for Wu et al.'s constant-coefficient case, and the entries of the lifting operator M_{x^p} that the same section uses for a variable coefficient.

The collocation residual at $N = 4$. For Wu et al.’s stiff pilot equation [1], $f'' + 5f' + 400f = 0$, Equation (3.4) gives $A = G^{\top 2} + 5G^{\top} + 400I$. At $N = 4$, using $G^{\top}$ (Appendix B),

$$A = \begin{pmatrix} 400 & 0 & 0 & 0 \\ 5\sqrt{2} & 400 & 0 & 0 \\ 4\sqrt{2} & 20 & 400 & 0 \\ 15\sqrt{2} & 24 & 30 & 400 \end{pmatrix}, \quad (\text{C.1})$$

verified directly against `pihm.chebyshev_derivative_matrix`. A is triangular here only because this equation has no odd-order term beyond f' at even N ; the off-diagonal entries sit three orders of magnitude below the diagonal, a large-coefficient conditioning problem stacked on top of G ’s own density.

The lifting operator M_{x^p} . M_{x^p} is exact and *rectangular*: multiplying a degree- $(N-1)$ model by x^p can raise its degree by as much as p , so the result is represented one register wider, $2N \times N$, with no truncation. At $n = 1$ ($N = 2$, so $p \in \{1, 2\}$),

$$M_{x^1} = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{2} \\ 0 & 0 \end{pmatrix}, \quad M_{x^2} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{3}{2\sqrt{2}} \\ \frac{1}{2} & 0 \\ 0 & \frac{1}{2\sqrt{2}} \end{pmatrix}, \quad (\text{C.2})$$

verified directly against `pihm.multiplication_matrix`. The general entry follows the same product-to-sum identity conjugated into the normalised bases of $|\tau(x)\rangle_n$ and $|\tau(x)\rangle_{n+1}$ that Section 3.2 states in closed form.

D The Published Encoding Circuit and Its Error Bound

Section 3.3 describes, at the level a reader needs to follow the argument, the circuit Wu et al. use to load $G^{\top}$ ’s column-linear entries as a rotation amplitude [1]. This appendix gives the mechanism in full, together with a rigorous bound on its entrywise error that this thesis derives and verifies – the source paper states only that the error is “negligible” and decreases “exponentially” with n , without a closed form.

The mechanism. A ladder of n singly-controlled R_y rotations acts on one flag qubit, initialised to $|1\rangle$: bit m of the (register-arithmetic-shifted) column index controls $R_y(\theta_m)$,

$$\theta_m = 2 \arcsin(-2^m \beta), \quad m = 0, \dots, n-1, \quad (\text{D.1})$$

for a small real coefficient β . Because rotations on the same qubit compose by exact angle addition ($R_y(a)R_y(b) = R_y(a+b)$, an $SO(2)$ identity), the net rotation for a column index p

with bit set S is $R_y(\Theta(p))$, $\Theta(p) = \sum_{m \in S} \theta_m$, giving the flag qubit's $|0\rangle$ -amplitude

$$\text{raw}(p) = -\sin(\Theta(p)/2) = \sin\left(\sum_{m \in S} \arcsin(2^m \beta)\right). \quad (\text{D.2})$$

The row-0 correction (entries $\sqrt{2}p$ in place of $2p$) is a second, independently accumulated ladder on the same flag qubit, with per-bit coefficient $\beta(1/\sqrt{2} - 1)$, controlled on the same “this is row 0” flag the main construction already computes.

Why this is only approximate. $\text{raw}(p)$ in Equation (D.2) would equal the desired linear value βp exactly only if $\sin\left(\sum_m \arcsin(x_m)\right) = \sum_m x_m$, which does not hold in general – $\sin$ of a sum is not the sum of sines. The discrepancy is a genuine, quantifiable small-angle error.

Proposition 1 (Entrywise error of the ladder construction). *Let $J_3(p) := \sum_{m \in S} (2^m)^3$ for column index p with bit set S . Then*

$$|\text{raw}(p) - \beta p| \leq \frac{\beta^3}{6} p^3 + O(\beta^5 N^5). \quad (\text{D.3})$$

Proof. Expand both non-linearities to leading order in the small parameter β : $\arcsin(y) = y + y^3/6 + O(y^5)$ and $\sin(x) = x - x^3/6 + O(x^5)$. Summing the first expansion over the fired bits and substituting into the second,

$$\sum_{m \in S} \arcsin(2^m \beta) = \beta p + \frac{\beta^3}{6} J_3(p) + O(\beta^5 N^5), \quad (\text{D.4})$$

$$\begin{aligned} \sin\left(\sum_{m \in S} \arcsin(2^m \beta)\right) &= \sin(\beta p) + \cos(\beta p) \cdot \frac{\beta^3}{6} J_3(p) + O(\beta^5 N^5) \\ &= \beta p + \frac{\beta^3}{6} (J_3(p) - p^3) + O(\beta^5 N^5). \end{aligned} \quad (\text{D.5})$$

Since $p = \sum_{m \in S} 2^m$ and every $2^m \leq p$, the multinomial expansion of p^3 term-by-term dominates $J_3(p)$, with equality only when S is a single bit (p itself a power of two): $J_3(p) \leq p^3$. Taking the worst case $p = N - 1$ gives Equation (D.3). $\square$

This is a leading-order bound, valid once the largest single argument, $2^{n-1}\beta$, is small and well inside $\arcsin$'s domain – true from $n = 2$ onward at the default $\beta = 2/\|G^\top\|_S$ (below), but not at $n = 1$, where $\beta \cdot 2^0 = \beta = 1$ exactly and the bound is not yet meaningful.

Verified scaling. With $\beta = 2/\|G^\top\|_S$, the S -norm subnormalisation Wu et al. define for this construction [1], $\|G^\top\|_S$ was measured directly against `pihm.chebyshev_derivative_matrix` for $n = 1$ through 8 and grows empirically as $\|G^\top\|_S \approx N^4/3$. Rescaling Equation (D.3) back to the units of $G^\top$'s own entries (multiplying by the construction's own subnormalisation, $(N/2) \cdot (2/\beta)$) gives the entrywise error bound reported in Table 1, verified against the measured error of the reconstructed circuit at every n shown.

n	N	$\ G^\top\ _S$	Entrywise bound ϵ_G
2	4	6.15×10^1	1.9×10^{-2}
3	8	1.18×10^3	1.3×10^{-3}
4	16	2.04×10^4	8.7×10^{-5}
5	32	3.38×10^5	5.6×10^{-6}
6	64	5.50×10^6	3.5×10^{-7}
7	128	8.88×10^7	2.2×10^{-8}
8	256	1.43×10^9	1.4×10^{-9}

Table 1: **The derived entrywise error bound ϵ_G , $n = 2$ to 8.** Each additional qubit shrinks the bound by roughly a further order of magnitude – substantially faster than N^{-1} , confirming Wu et al.’s own qualitative claim of exponential decrease and giving it, here, a closed form and a verified value.

Resource count. The primary and correction ladders act on one flag qubit, controlled jointly on the (register-arithmetic-shifted) column-index register and a single “row zero” flag; the shift itself is computed by a ripple-carry adder over the n -qubit system register and uncomputed afterwards, so no ancilla are retained beyond bookkeeping. The reconstruction built for this thesis uses $n + 4$ ancilla in total (a borrow flag, the value-load flag, an exclusion flag and a was-zero flag, alongside the n -qubit offset-selector register); an exact, non-approximating alternative – a full register-arithmetic data load with no ladder at all, used only to generate the reference values Table 1 is checked against – costs $n + 5$ ancilla and depth $\Theta(2^n)$ in place of the ladder’s $\Theta(n)$. Both are within one qubit of the $n + 3$ ancilla Wu et al. report for their own circuit [1]; the gap is attributable to this reconstruction keeping two flag qubits live rather than one combined flag, and was not pursued further, since it does not change any scaling claim in Section 3.5.

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